

**National University of Computer and Emerging Sciences, Lahore Campus**

|                    |                                          |                     |                  |
|--------------------|------------------------------------------|---------------------|------------------|
| <b>Course:</b>     | <b>Design and Analysis of Algorithms</b> | <b>Course Code:</b> | <b>CS2009</b>    |
| <b>Program:</b>    | <b>BS(Data Science)</b>                  | <b>Semester:</b>    | <b>Fall 2023</b> |
| <b>Duration:</b>   | <b>20 Minutes</b>                        | <b>Total Marks:</b> | <b>14</b>        |
| <b>Paper Date:</b> | <b>14-Sep-2022</b>                       | <b>Weight</b>       | <b>%</b>         |
| <b>Section:</b>    | <b>5A v1</b>                             | <b>Page(s):</b>     | <b>2</b>         |
| <b>Exam:</b>       | <b>Quiz 1</b>                            |                     |                  |

**Q(1) Version 1** Use a recursion tree to determine a good asymptotic upper bound on following recurrence (5 Marks)

$$T(n) = T(n/3) + O(n)^3$$

$$T(n) = n^3 + (n/3)^3 + (n/9)^3 + \dots$$

$\leq n^3$  (geometric series with  $x = 1/27$ ,  $1/(1-1/27)$  see Appendix A of book for formula of geometric series that goes till infinity)

$$O(n^3)$$

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**Q(1) Version 2**

$$T(n) = 3T(3n/4) + O(1)$$

Height :  $\log_{4/3} n$

Since it is constant, we add the nodes only.

At level 1, we have 1 node

At level 2, we have 3 nodes

At level 3, we have 9 nodes

At level 2, we have 27 nodes

Geometric Series can be applied

Common ratio is 3

Answer :  $O(n^{3.8})$

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**(Q2) Version 1** True / False: Justify your answer (3+3+3 Marks) (1 Mark for correct answer, 2 marks for justification, prove using constants)

a)  $5^{n+9} = O(5^n)$

$$5^{n+9} \leq c 5^n$$

$$5^n 5^9 \leq c 5^n$$

$$5^9 \leq c$$

True for  $c = 5^9$  and  $n_0 = 1$

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b)  $4^{7n} = O(4^n)$

$$4^{7n} \leq c 4^n$$

$$4^{6n+n} \leq c 4^n$$

$$4^{6n} 4^n \leq c 4^n$$

$$4^{6n} \leq c$$

This is a contradiction. There is no constant that is always greater than  $4^{6n}$  so it is False

c)  $27^n = \Theta(9^n)$

$$c_1 9^n \leq 27^n \leq c_2 9^n$$

$$c_1 3^{2n} \leq 3^{3n} \leq c_2 3^{2n}$$

$$c_1 3^{2n} \leq 3^{2n+n} \leq c_2 3^{2n}$$

$$c_1 3^{2n} \leq 3^{2n} 3^n \leq c_2 3^{2n}$$

$$c_1 \leq 3^n \leq c_2$$

False as we cannot find  $c_2$  that is always greater than  $3^n$

**(Q2) Version 2**

Q.2

1a)  $n^2 \log_6 n = O(n^3 \log_{10} n)$

$$n^2 \log_6 n \leq c n^3 \log_{10} n$$

$$\log_6 n \leq c (n \log_{10} n)$$

True

$$\frac{\log_6 n}{\log_{10} n} \leq c \cdot n$$

We can find a ~~for~~ constant.

1b)  $3^{5n} = \Theta(3^{2n})$

$$3^{5n} \leq c 3^{2n}$$

False

$$(3^{2n})(3^{3n}) \leq c 3^{2n}$$

$$3^{3n} \leq c$$

doesn't hold

1c)  $125^n = \Theta(25^n)$

$$c_1 25^n \leq 125^n \leq c_2 25^n$$

$$c_1 (5^2)^n \leq (5^3)^n \leq c_2 (5^2)^n$$

$$c_1 \leq 5^n \leq c_2$$

True and False

False